SIMULTANEOUS GO VIA QUANTUM COLLAPSE

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Abstract. We construct a symmetric, simultaneous, stateful game ***SGo***, which is in a certain mathematical sense a symmetrization of the classical board game Go. ***SGo*** is in some ways a simpler game than Go, as Komi and Ko rules are removed. On the other hand the states are more complex, because there are "entanglement phenomena", an idea inspired by quantum mechanics. However there are no dice rolls, instead there is a certain deterministic "quantum state" reduction, so that state evolution is deterministic. Using the argument of Nash, we show that ***SGo*** has an mixed equilibrium strategy to draw on average. One open problem is whether chess similarly has a symmetrization in our sense.

1. Introduction

The game Go is an ancient and interesting game of local to global geometric principles. Like all sequential games it suffers from time asymmetry between the players Black and White. In Go this is addressed by an ad hoc rule called Komi: whereby the player White which moves second is given a specific point advantage. For Go, time asymmetry also results in certain local pathologies in game states, for example Ko situation, where after a capture of a stone the opposing player may immediately recapture, but is forbidden by an extra ad-hock rule, to avoid an infinite loop situation.

At the same time, precisely because of its local to global nature Go is "almost time symmetric". We make this mathematically precise by constructing a symmetric simultaneous (stateful) game ***SGo*** that is in one mathematical sense a symmetrization of Go. ***SGo*** is a deterministic game, which is totally playable on a standard Go board, although we need a third type of stones that we call red. ***SGo*** is in some ways a simpler game than Go as it loses the Komi and Ko rules, and it has just two rules stone placement and capture (aside from end of game rule), but the latter and game states are now more complicated, as we have to deterministically resolve simultaneous action. The idea for ***SGo*** is inspired by quantum mechanics.

One open problem is whether chess similarly has a symmetrization in our sense. Using the famous idea of Nash in [1], we show that ***SGo*** has a mixed equilibrium strategy to draw on average. That said as will be apparent from the argument, finding this strategy by a computational approach is totally impossible, due to the unreal size of the strategy space. We can also ask if ***SGo*** has a strategy to draw not just draw on average.

2. Game rules

The game is initialized with an empty standard Go board, with virtual horizontal/vertical lines whose intersections we just call intersections. There are two players that we call Black and White. As in Go, players Black and White move on a given turn by placing a black respectively white stone on an empty intersection on the board, but in *SGo* they do this on the same turn without knowledge of the others move on this turn. This requires modifying the placement and capture rules of Go, while removing the Komi rule, and Ko rules. We also need to introduce a third type of stones called red, which may be informally interpreted as simultaneously white and black.

2.1. Some definitions. Given a board game state S, that is any collection of white, black, red stones on the board let $G(S)$ denote the graph with vertices stones, and edges between vertices whenever the corresponding stones are horizontally or vertically adjacent. (Meaning they are adjacent and on the same horizontal/vertical line.) From now on adjacency, always means this type of adjacency. In what follows the terms stones and vertices may be used synonymously.

A colored component is a connected component of the full (a.k.a. induced) subgraph of $G(S)$ whose vertices are either all the black, or all the white stones. We call this a white, respectively black component. (Full subgraph means that all the edges of the original graph, between any pair of vertices of the subgraph, are included.)

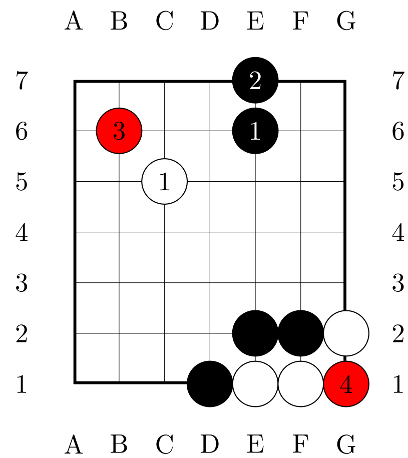
As in classical Go, a connected component C has no liberties if it cannot be extended (as a connected component) by placing a stone on the board. We will also call C a trapped component. We define stones adjacent to C, as those stones whose corresponding vertices are connected by an edge to C in $G(S)$.

Given a colored component C, A white/black stone placed at an intersection is called a trapping if it is adjacent to a trapped colored component of opposite color.

2.2. Stone placement resolution rule of *SGo*. If the players move to the same intersection on the board we place a stone there whose color resolves according to the following. If either a white or black stone placed at the intersection is trapping, then a red stone is placed at the intersection. We call this a trapping entanglement. Otherwise, if the move is not a trapping trapping entanglement we have the following:

(1) If the intersection is adjacent to a black stone, and neither a white stone nor red stone, a black stone is placed at the intersection.
(2) If the intersection is adjacent to a white stone, and neither a white stone nor red stone a white stone is placed at the intersection.
(3) Otherwise, a red stone is placed.
(4) If the stone placed by Black/White is not trapping, but is contained in a trapped colored component, it is ruled a suicide, and is an illegal move.

To the right is an example of placement resolution. Note that the stone placed on move 4 does not resolve to white, as it is trapping.



2.3. Capture rule of *SGo*. After both White and Black move on current turn numbered n, using the stone placement rule above, we get some board state S.

The stage 1 capture rule is then invoked on this turn as follows. Let C be a colored component with no liberties of the board state S, (the stones of C may have been placed on this turn.) Suppose that:

(1) There is no red stone adjacent to C placed on turn n.
(2) Some stone adjacent to C was placed on turn n.

Then we call C trapped on turn n, the stones of C are also called trapped on turn n turn. Suppose that no stone adjacent to C was trapped on this turn. Then C is resolved as captured.

In the case C is captured, supposing C is white:

- The stones of C are removed as prisoners of Black as in Go.
- Any red stones adjacent to C resolve as black. (Concretely are replaced by black stones.)

If C is black the process is reflected. The above stage 1 capture rule is invoked recursively until no captured colored components remain. We call this stage one reduction and the resulting board state will be denoted by $R_n(S)$.

Lemma 1. The stage one reduction is well defined, that is independent of the order in which killed components are captured.

Proof. Suppose we have two colored components C_0, C_1 captured on the given turn. By assumptions no stones in C_1 can be adjacent to stones of C_0, and vice versa. Hence applying capture rule to the group C_0 and then to group C_1, results in the same board state as first applying the capture rule to the group C_1 and then the group C_0. The result clearly follows. □

If there are no colored components trapped on the turn $n-1$ or lower, the turn is resolved. Otherwise, we recursively define higher stage reduction as follows.

Define stage k reduction by the condition that the board state after stage k reduction denoted by $R^k(S)$ satisfies:

$$R^1(S) = R_n(S).$$
$$R^k(S) = R_{n-k+1}(R^{k-1}(S)).$$

If there are no colored components trapped on the turn $n-k$ or lower, the turn is resolved after stage k reduction.

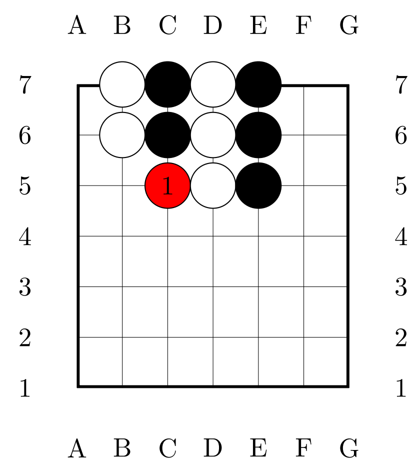
Remark 2. This may appear to be fairly complicated, but in practice the resolution of most turns is easy to work out mentally, we will give some examples of such "nested entanglement states" further ahead. Of course having a computer work out turn resolution automatically is very helpful.

2.4. End of the game. As in Go, *SGo* game ends after both players pass on the same turn. At this point the winner is decided as in Go by territory and prisoner count. White's territory boundary consists of connected components of white/red stones. Blacks territory boundary consists of connected components of black/red stones. Red stones are never counted as prisoners. A draw is now possible although compared to chess it might be much less likely, (on a typical 13 ×13 or 19 × 19 go board). See Section 3.2 for an example.

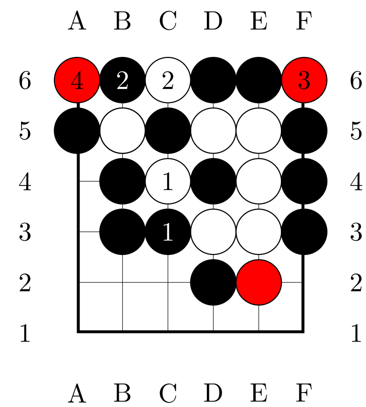
3. Examples

We give here some examples. In most of these example, the players need only play the position with a pure strategy in game theory language. In more marginal, complex positions, the players may need mixed strategies. See however Question 2.

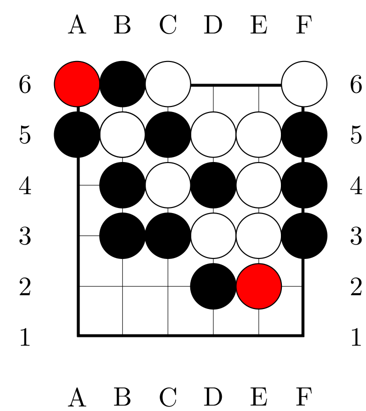
On move 1 both Black and White move to C5. The left pair of black stones will be captured next turn.



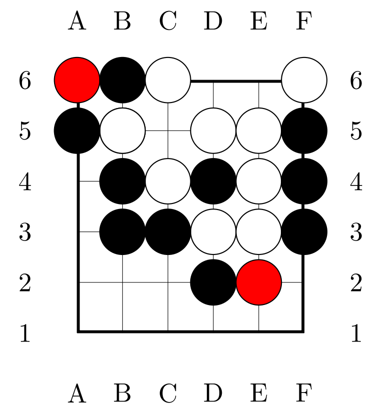
In the position on the right, on turn 1, White attempts capture at C4, while Black attempts capture at C3. This gives an example of a “nested entangled state”. A possible continuation is indicated.



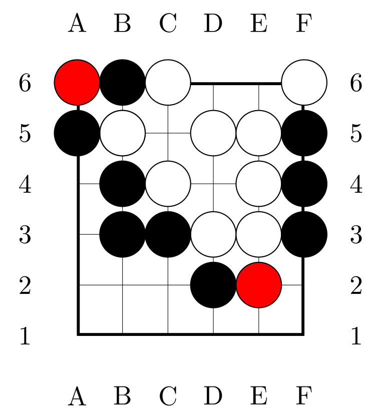
At the end of turn 4, stage one resolution is depicted on the right.



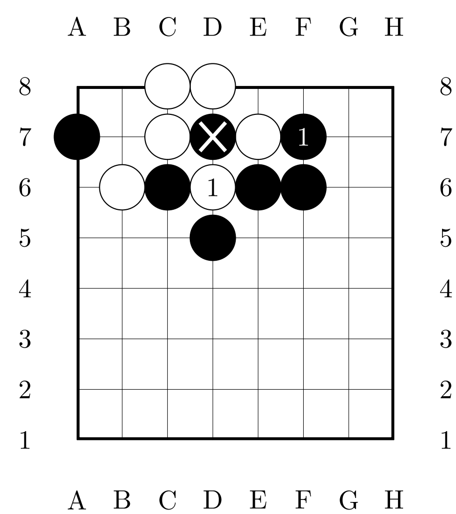
Stage two resolution is depicted on the right.



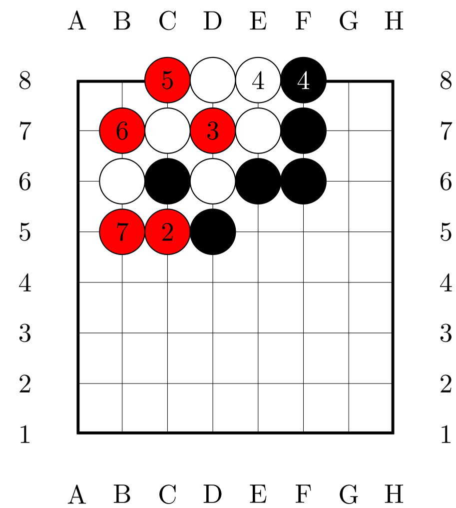
Stage three and final state resolution is depicted on the right.



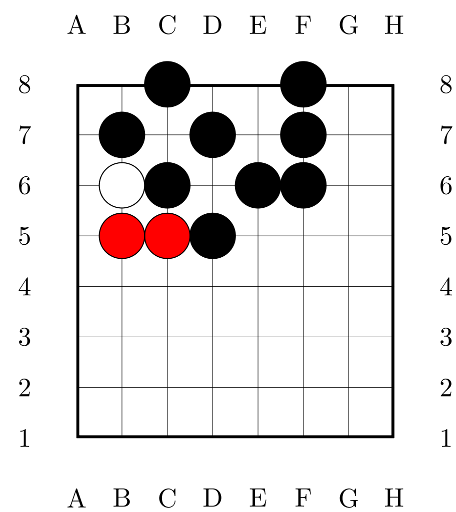
Ko situation. Recall, that we don't have a Ko rule, but it is now clearly unnecessary. See example on the right. White's D6 is a capture. Note that this is already implicitly in the rules set, because the white stone at D6 is not trapped on this turn. By the rules, to be trapped on this turn a stone cannot be created on this turn. This behavior is chosen for backwards compatibility with Go, and for the purpose of Theorem 1 in particular.



Here is a possible continuation.

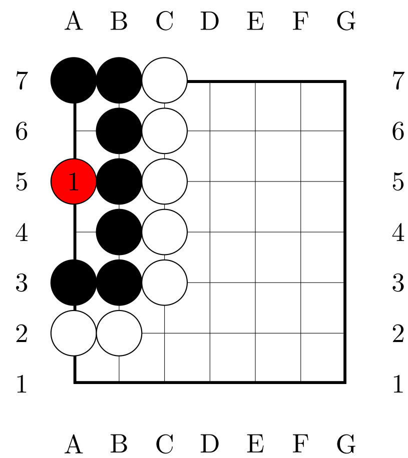


On turn 7 the position resolves as follows.

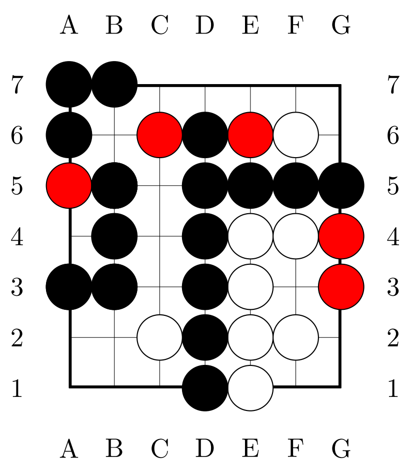


3.1. Life and death. Life and death is positionally very similar to Go, here is an example.

The Black group is alive, that is impossible to kill.



3.2. End game position. Both players pass and the game ends. Assuming there are no prisoners. White has 4 points, Black has 19 points.



4. SGo as a simultaneazation of Go

We give here a general definition of symmetrization of a sequential deterministic game, and show that *SGo* is a symmetrization of *Go*. The result is not unsurprising, for example chess likely does not have a symmetrization in our sense, although formal verification of this is not obvious. First we give some basic definitions. We will treat (finite) deterministic games as certain types of (finite) deterministic automata.

Definition 1. A (finite) a deterministic game G with 2 players P_0, P_1 or Black, White, consists of:

- A (finite) deterministic automaton G, whose set of states is a set $S(G)$ called the set of game states of G.
- Nonempty subsets $W_0(G), W_1(G), D(G) \subset S(G)$ of final (accepting) states with empty pairwise intersection, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in S(G)$, understood as the state in which the game is initialized.
- A $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \sqcup M_1(G)$: interpreted as possible moves of Black and White.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \to S(G).$$

We recursively define:

$$E(s, m_1, \ldots, m_k) = E(E(s, m_1, \ldots, m_{k-1}), m_k).$$

From now on we restrict to finite deterministic games for simplicity, and just call them games.

4.1. Sequential games.

Definition 2. A game G is called sequential if the following conditions are satisfied:

(1) The finite set $S(G)$ has a distinguished representation as

$$S(G) = B(G) \times \mathbb{Z}_2,$$

where $B(G)$ is some set. ($B(G)$ may be interpreted as the set of "board states").

(2) The function E has the properties: if $s \in B(G) \times \{0\}$, and $m \in M_1(G)$ then $E(s,m)$ is undefined. Likewise, if $s \in B(G) \times \{1\}$, and $m \in M_0(G)$ then $E(s,m)$ is undefined.

(3) If $s \in B(G) \times \{0\}$, and $E(s,m)$ is defined then $E(s,m) \in B(G) \times \{1\}$. Likewise, if $s \in B(G) \times \{1\}$, and $E(s,m)$ is defined then $E(s,m) \in B(G) \times \{0\}$.

4.2. Simultaneous games. What we describe here can also be understood as certain kinds of stochastic 2-player with deterministic evolution, but the language of automata is a bit simpler here, and it allows us to treat everything on the same footing.

Definition 3. A game G as above is called simultaneous if the following conditions are satisfied:

(1) The finite set $S(G)$ has a distinguished representation as

$$S(G) = B(G) \sqcup B(G) \times M_0(G) \sqcup B(G) \times M_1(G),$$

where $B(G)$ is some set.

(2) $q_0 \in B(G)$ and $W_0(G), W_1(G), D_G \subset B(G)$.

(3) The function E has the properties: if $s \in B(G) \subset S(G)$, then $E(s,m) = (s,m)$ for all $m \in M(G)$ s.t. $E(s,m)$ is defined. If $s \in B(G) \times M_0(G)$, and $m \in M_1(G)$ then $E(s,m) \in B(G)$ (or undefined). If $s \in B(G) \times M_1(G)$, and $m \in M_0(G)$ then $E(s,m) \in B(G)$ (or undefined). E is undefined in all other cases.

(4) Suppose that $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, $s \in B(G)$, $E(s, m_0)$ is defined and $E(s, m_1)$ is defined then $E(s, m_0, m_1)$ is defined.

(5) For s, m_0, m_1 as in the previous property, if $E(s, m_0, m_1)$ is defined then so is $E(s, m_1, m_0)$ and
$$E(s, m_0, m_1) = E(s, m_1, m_0).$$

The above notion is more interesting when G also satisfies a certain symmetry between the two players.

Definition 4. We say that a simultaneous game G is symmetric if the following conditions are satisfied.

(1) There is a bijection $\mathcal{S}: M(G) \to M(G)$, taking $M_0(G)$ onto $M_1(G)$.
(2) There is a bijection $R: B(G) \to B(G)$ s.t. for all $s \in B(G)$, all $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, and for $\mathcal{S}$ as above we have:
   - $E(R(s), \mathcal{S}(m_0), \mathcal{S}(m_1)) = R(E(s, m_0, m_1))$, whenever both sides are defined. Furthermore, if one side is defined then so is the other.
   - R also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

4.3. Simultanezation and symmetrization. Let G be a sequential game. Set $pr_{B(G)}: S(G) \to B(G)$ to be the natural projection map. Define $P: B(G) \times M_0(G) \times M_1(G) \to 2^{B(G)}$ by the following conditions.
$$P(s, m_0, m_1) := pr_{B(G)} \circ E_G((s,0), m_0, m_1) \sqcup pr_{B(G)} \circ E_G((s,1), m_1, m_0),$$
assuming the right hand side is defined. Otherwise if $E_G((s,0), m_0, m_1)$ is defined but $E_G((s,1), m_1, m_0)$ is undefined set
$$P(s, m_0, m_1) := pr_{B(G)} \circ E_G((s,0), m_0, m_1) \sqcup pr_{B(G)} \circ E_G((s,1), m_1).$$
Otherwise if $E_G((s,1), m_1, m_0)$ is defined but $E_G((s,0), m_0, m_1)$ is undefined set
$$P(s, m_0, m_1) := pr_{B(G)} \circ E_G((s,0), m_0) \sqcup pr_{B(G)} \circ E_G((s,1), m_1, m_0).$$
Otherwise if $E_G((s,1), m_1, m_0)$ and $E_G((s,0), m_0, m_1)$ are both undefined set
$$P(s, m_0, m_1) := pr_{B(G)} \circ E_G((s,0), m_0) \sqcup pr_{B(G)} \circ E_G((s,1), m_1),$$
provided the right hand side is defined. In all other cases $P(s, m_0, m_1)$ is undefined.

Definition 5. Let G be a sequential game. Its formal simultanezation is a simultaneous game $Sim(G) = G'$ satisfying the following.

(1) $B(G') = 2^{B(G)}$, with the initial state $\{q_0\} \in B(G')$.
(2) $M(G') = M(G)$ as $\mathbb{Z}_2$ graded sets.
(3) Let $\mathfrak{s} \in B(G')$, and $m_0 \in M_0(G')$, $m_1 \in M_1(G')$. Then
$$E'(\mathfrak{s}, m_0, m_1) := \cup_{s \in \mathfrak{s}} \{P(s, m_0, m_1)\},$$
(undefined if some $P(s, m_0, m_1)$ is undefined).

This formal simultanezation has an impractically large number of states. This can be cut down in various ways, but usually this will not result in a symmetric game. So we make the following definition.

Definition 6. Let G_0, G_1 be games. We say that G_1 is a symmetrization of G_0 if the following is satisfied.

(1) G_1 is a symmetric simultaneous game.
(2) $M(G_1) = M(G_0)$ as $\mathbb{Z}_2$ graded sets.

(3) There is a map $\rho : B(G_1) \to B(Sim(G))$ s.t. for $s \in B(G_1)$ $E_{G_1}(s, m_0, m_1)$ is defined iff $E_{Sim(G)}(\rho(s), m_0, m_1)$ is defined, and in this case

$$\rho(E_{G_1}(s, m_0, m_1)) \subset E_{Sim(G)}(\rho(s), m_0, m_1).$$

(4) Finally, we have a property that to paraphrase says that the complexity of G_1 is comparable to that of G_0. There is a map $\phi : B(G_1) \to B(G_0)$ s.t. for $s \in B(G_1)$, $E_{G_1}(s, m)$ is defined if and only if $E_{G_0}(\phi(s), m)$ is defined.

Theorem 1. SGo is a symmetrization of Go.

Proof. We first outline the formal description $G = SGo$ as a simultaneous game, based on the description in Section 2. The space $M(G)$ is the same as Go, we suppose that $M_0(G)$ corresponds to moves of Black. The state space is:

$$S(G) = B(G) \sqcup B(G) \times M_0(G) \sqcup B(G) \times M_1(G),$$

where an element of $B(B)$ is a legal board position and its entire legal game history. By legal, we mean the history is obeying the rules of Section 2.

The evolution map E is then obviously defined. For example if $b \in B(G)$ and $m_0 \in M_0(G)$, $m_1 \in M_1(G)$ are legal moves then $E((b, m_0), m_1) \in B(G)$ is defined to be the resolution of the board position, using the placement and capture rules of Section 2, after Black's move m_0, and White's move m_1 are reflected on the board.

To see that G is symmetric, let $R : B(G) \to B(G)$ be the map that interchanges the black and white colors of the stones of the board states, leaving red stones invariant. And let $\mathcal{S} : M(G) \to M(G)$ interchange black and white moves, i.e. we have a natural isomorphism $M_0(G) \to M_1(G)$, which then induces $\mathcal{S}$. Then clearly the conditions of Definition 4 will be satisfied.

Property 3 essentially follows by construction of SGo. The map ρ corresponds to taking resolutions of a SGo board state s into classical Go board states. For example, assume that there is no entanglement that is colored components C_0, C_1 of opposing colors that were trapped on some turn n (and remain trapped) but with some stones of C_0 adjacent to stones of C_1. Then for each red stone in an SGo board there are two resolutions corresponding to replacing it by a black/white stone, and then reducing to a legal Go board position, by performing capture if necessary.

When there is entanglement, first resolve all red stones to get some set $\{s_0, \ldots, s_n\}$ of SGo positions with no red stones. For each s_i, we resolve entanglement in the reverse order of turn on which it was created, with the most recent turn first. The resolution is by giving Black or White the initiative (i.e. they acted first). Then descend to the previous turn and recurse. At the end of recursion, for each s_i we then get another collection of classical Go states τ^i. Finally, we set $\rho(s) = \bigcup_i \tau^i$.

To verify the final property, define $\phi : B(G) \to B(Go)$ by the following conditions. $\phi(s)$ is the Go board state obtained by replacing every stone on the board by a black stone. It is then immediate that Property 4 is satisfied. □

Question 1. Does chess have a symmetrization?

Of course chess has the formal simultanezation $Sim(chess)$, and this moreover can be made into a symmetric simultaneous game after properly defining final states in $Sim(chess)$. But Property 4 will not be satisfied. And non-locality of chess makes it difficult to reduce $Sim(chess)$ to something that does satisfy Property 4.

4.4. Solvability. A theorem of Zermelo [2] says roughly that in every sequential game, either Black or White has a winning strategy or there is a draw strategy. Obviously *SGo* cannot have a winning strategy.

Question 2. Does SGo have a strategy to draw (meaning a pure Nash equilibrium strategy to draw)?

We point out that there are trivial examples of simultaneous games with a strategy to draw. For a very trivial example, take any finite simultaneous game G and reset $D(G) = B(G) - q_0$, $W_0(G) = \emptyset$, $W_1(G) = \emptyset$. Of course we have simultaneous games with no strategy to draw, the classical example is rock-paper-scissors. But the latter has a strategy to draw on average. The latter being a very trivial case of the existence of the Von-Neumann, Nash equilibrium for "matrix payoff games", [1]. SGo is not a matrix payoff game. However, Nash's proof of the equilibrium theorem readily extends to this case. In general:

Theorem 2 (Nash). Assigning any payoff for a win and draw, a finite simultaneous game G in the sense above has a Nash equilibrium.

Proof. For $s \in B(G)$ set

$$M_i(s) = \{m \in M_i(G) \,|\, E(s,m) \text{ is defined}\}.$$

The space of strategies of player P_i is then:

$$\mathcal{P}_i = \prod_{s \in B(G)} \Delta(M_i(s)),$$

where $\Delta(M_i(s))$ is the space of probability distributions over the finite set $M_i(s)$. Since $B(G)$ is also finite, the joint strategy space $\mathcal{P}_0 \times \mathcal{P}_1$ is a compact convex subspace of $\mathbb{R}^N$, for some very large N.

The argument of Nash [1] using Kakutani's fixed point theorem then readily extends to get existence of an equilibrium point. □

Corollary 1. A symmetric simultaneous game has a strategy to draw on average, and in particular *SGo* has such a strategy.

Proof. And assign 1 for a win, -1 for a loss and 0 to a draw. Then it is enough to show that for any Nash equilibrium strategy the expected payoff is 0 for both players.

Suppose otherwise, and suppose without loss of generality that $(\mathcal{S}_0, \mathcal{S}_1)$ is an equilibrium joint strategy where Black has a positive expected payoff.

Let $R, \mathcal{S}$ be the symmetry operators as in the proof of Theorem 1. Applying these operators to the strategy $\mathcal{S}$ we get that there is an equilibrium joint strategy $(\mathcal{S}'_0, \mathcal{S}'_1)$ where White has positive expected payoff. But this is absurd. □

References

[1] J. F. j. Nash, Equilibrium points in n-person games, Proc. Natl. Acad. Sci. USA, 36 (1950), pp. 48–49.

[2] U. Schwalbe and P. Walker, Zermelo and the early history of game theory, http://www.math.harvard.edu/~elkies/FS23j.03/zermelo.pdf.